



# **Device Credit Monitoring System**

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## Tab Explanation and Step-by-Step User Guide

This guide explains each tab in the credit system, what information appears there, who normally uses it, and the exact steps to complete common work. It is written for admin, Biomed, Accounts, and viewer users.

Reference style: module purpose, information displayed, main actions, workflow, and controls. This version goes deeper into credit balance, billing, notifications, reports, and permissions.

## System Flow

**End-to-end process:** Sign in -> Review Dashboard -> Import or select device -> Record credit activity -> Balance recalculated -> Accounts updates invoice/payment -> Schedule and reports update -> Alerts sent when required

## Sidebar Structure Overview

This is the practical navigation map for the left sidebar. Use it to understand which tab opens which function, and which actions normally happen inside each tab.

### Left Sidebar

- |-- **Dashboard**
  - | |-- Date range filter
  - | |-- KPI review
  - | |-- Trends, rankings, low-balance alerts, notifications
- |-- **Credit Monitoring**
  - | |-- **Credit Movement**
    - | | |-- Search and filter transactions
    - | | |-- Export Excel / Export PDF
    - | | |-- New Transaction
    - | | |-- Edit Transaction
    - | | |-- Update Payment
    - | | |-- View History
    - | | |-- WhatsApp credit summary
    - | | |-- Delete Transaction, primary admin only
  - | |-- **Purchase Schedule**
    - | | |-- Select device
    - | | |-- View rolling schedule
    - | | |-- Check invoice/payment status for cycles
- |-- **Clients**
  - | |-- Search clients
  - | |-- Generate Client Report
  - | |-- View Devices for selected client
- |-- **Devices**
  - | |-- Search/filter by client and status
  - | |-- Monitor active, low-balance, and unassigned devices
  - | |-- Edit low-balance threshold, Admin/Biomed
  - | |-- Start New Transaction / import Biomed device
- |-- **Revenue Report, Admin/Accounts**
  - | |-- Select reporting period

- | |-- Generate revenue statement
- | |-- Review outstanding quotations
- | `-- Print / Save PDF
- |-- Users, Admin**
  - | |-- Search/filter users
  - | |-- Add User
  - | |-- Edit User
  - | `-- Activate / Deactivate User
- |-- Settings, Admin**
  - | |-- **General**
    - | | |-- Default low-balance threshold
    - | | `-- Schedule reminder days
  - | |-- **Activities**
    - | | |-- Add Activity
    - | | |-- Edit Activity
    - | | `-- Deactivate Activity
  - | `-- **Email Notifications, admin@admin.com only**
    - | |-- Add Recipient
    - | |-- Edit Recipient
    - | `-- Delete Recipient
- `-- Profile Menu**
  - |-- Profile
  - |-- Change Password
  - `-- Sign out

| Sidebar tab       | Sub pages/actions                                                                   | Access note                                                                     |
|-------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Dashboard         | Date range, KPIs, utilization/payment charts, alert lists, recent notifications.    | All roles.                                                                      |
| Credit Movement   | New, edit, payment update, history, WhatsApp summary, delete, Excel/PDF export.     | Create for Admin/Biomed; payment for Admin/Accounts; delete primary admin only. |
| Purchase Schedule | Device selector, rolling schedule, purchase/invoice/payment status.                 | All roles can view.                                                             |
| Clients           | Search, client report, view client devices.                                         | All roles can view reports.                                                     |
| Devices           | Search/filter, low-balance monitoring, threshold edit, new transaction/import path. | Threshold/import for Admin/Biomed.                                              |
| Revenue Report    | Reporting period, generated statement, outstanding quotations, print/save PDF.      | Admin/Accounts only.                                                            |
| Users             | Search/filter, add, edit, activate/deactivate.                                      | Admin only.                                                                     |
| Settings          | General, Activities, Email Notifications.                                           | Admin only; Email Notifications only admin@admin.com.                           |
| Profile Menu      | Profile, password change, sign out.                                                 | All roles.                                                                      |

| Main area         | Purpose                                                                                            | Typical users                        |
|-------------------|----------------------------------------------------------------------------------------------------|--------------------------------------|
| Dashboard         | High-level operating, finance, usage, and alert overview.                                          | All roles                            |
| Credit Movement   | Main transaction ledger for top-ups, utilization, billing, payment, history, and export.           | Admin, Biomed, Accounts, Viewer      |
| Purchase Schedule | Shows expected top-up cycles based on contract start date and top-up frequency.                    | All roles                            |
| Clients           | Client list, total assigned devices, total credits, and client report access.                      | All roles                            |
| Devices           | Device credit balance monitoring, low-balance thresholds, and Biomed device import/sync workflows. | All roles; edit for Admin/Biomed     |
| Revenue Report    | Finance report for sold credits, revenue, usage, and quotations awaiting invoice.                  | Admin, Accounts                      |
| Users             | Create, edit, activate, and deactivate staff accounts.                                             | Admin                                |
| Settings          | Configure thresholds, schedule reminders, activities, and email notification list.                 | Admin; email list only primary admin |
| Profile           | Review own account details and change password.                                                    | All roles                            |

## Role Access

Access is controlled by the role attached to the signed-in user. Some actions are also restricted to the primary administrator account.

| Role                | Can view                                                                                  | Can change                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Admin               | All operational, finance, report, user, and settings pages.                               | Create/edit transactions, update payment, import/sync devices, edit thresholds, manage users, manage settings. |
| Biomed              | Dashboard, Credit Movement, Purchase Schedule, Clients, Devices, Profile.                 | Create transactions, edit Biomed/operations fields, import/sync Biomed devices, edit device thresholds.        |
| Accounts            | Dashboard, Credit Movement, Purchase Schedule, Clients, Devices, Revenue Report, Profile. | Edit account fields, update payment status/date, add invoice details, upload device image, add notes.          |
| Viewer              | Dashboard, Credit Movement, Purchase Schedule, Clients, Devices, Profile.                 | View, filter, export, and review history only.                                                                 |
| Primary admin email | Same as Admin.                                                                            | Delete credit transactions and manage email-notification recipients when email is admin@admin.com.             |

## 1. Dashboard

**Purpose:** The Dashboard is the daily overview page. It helps users identify credit usage, payment performance, devices with no usage, and devices approaching low credit before opening the detailed tabs.

### Information Displayed

- Active Contracts, Total Clients, Assigned Devices, Low Balance Alerts, Pending Revenue, and Utilization.
- Utilization Trends chart showing credits tracked across the selected period.
- Payment Mix chart showing amount by payment status.
- Top 10 Customers by Usage.

- Credits Sold by Salesperson for top-up activity.
- Most Active Devices and Devices with Zero Usage.
- Devices Approaching Low Credit with balance, threshold, and status.
- Recent unread notifications for low balance, payment received, schedule reminder, or general notices.

## How To Review Daily Risk

1. Open Dashboard after signing in.
2. Check Low Balance Alerts first. A non-zero number means at least one active device is at or below its threshold.
3. Review Devices Approaching Low Credit to see the client, device serial number, current balance, and threshold.
4. Open the Devices tab if the threshold or device list needs more detail.
5. Open Credit Movement if the balance looks wrong and transaction history needs to be checked.

## How To Change The Dashboard Period

1. Click the Date range control near the Dashboard title.
2. Choose a preset such as This month, Last 30 days, Last month, or All time.
3. For a custom period, select Custom, pick the start date, then pick the end date.
4. Click Update. The dashboard KPIs, charts, and ranking tables reload for that date range.

## 2. Credit Movement

**Purpose:** Credit Movement is the main ledger for every credit transaction. It joins Biomed operational information with Accounts billing and payment information in one record.

### Information Displayed

| Column group        | Fields shown                                                                                                               |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| Biomed & Operations | Client, Date From, Date To, Credit Before, Credit After, Total Utilization, Device SN, Activity, Sales Person, Credit Qty. |
| Account Department  | Total Credit to Bill, Price/Credit, Discount, Total Charge, Invoice Ref., Payment Status, Payment Date.                    |
| Actions             | WhatsApp summary, History, Edit, Delete for primary admin, and Update Payment when an invoice exists.                      |

### Balance Logic

When a transaction is saved, the system calculates: new balance = credit balance before + total credit quantity - total utilization.

- Top-up and free-credit activities increase the balance through Credit Qty.
- Usage or loss activities reduce the balance through Total Utilization or negative credit quantity depending on the activity setup.
- The saved transaction keeps both Credit Before and Credit After so the ledger can be audited later.

- After create, edit, payment update, or delete, the schedule/report views use the updated transaction data.
- If the final balance is at or below the device threshold, low-balance notification logic can alert active users and email recipients.

## **How To Search And Filter Transactions**

1. Open Credit Monitoring > Credit Movement.
2. Use the search box to find a client, device serial number, or invoice reference.
3. Use Client, Activity, and Payment Status filters to narrow the table.
4. Use Date From and Date To when you only want transactions recorded within a period.
5. Click Clear to return to the full transaction list.

## **How To Create A New Credit Transaction**

1. Open Credit Movement and click New Transaction. This is available to Admin and Biomed users.
2. Select the device. The dropdown can show existing system devices and can load Biomed devices for import when available.
3. Confirm the CRM Customer Details: client name, department or branch, address, contact, PIC, and telephone. If a device is linked to CRM, these fields are populated from CRM data.
4. Review Contract Details: business model, duration, top-up frequency, start date, minimum commitment, and low-balance threshold for a new device.
5. In Credit Activity, choose the activity type. For top-ups, choose a top-up package or custom amount, select the salesperson, and enter Credit Qty. For utilization, enter Utilization.
6. Use Add Activity when one transaction contains multiple activities. The system allows only one top-up activity per transaction.
7. Enter Date From and Date To for the period covered by the transaction.
8. Review Credit Before and Total Credit After Activity. These are calculated fields.
9. If you have Accounts access, complete Total Credit to Bill, Price/Credit, Discount, Invoice Ref. No., Payment Status, Payment Date, Device Picture, and Notes.
10. Click Save Transaction. The system creates history, updates the contract balance, updates price per credit, and checks low-balance notification rules.

## **How To Edit A Transaction**

1. Find the transaction in Credit Movement.
2. Click the Edit icon.
3. Admin can edit both Biomed and Accounts fields. Biomed can edit operations fields. Accounts can edit billing/payment fields.
4. Make the required changes and review the recalculated balance or charge.
5. Click Update Transaction. A transaction history entry is created with before-and-after snapshots.

## **How To Update Payment Only**

1. Find a transaction that has an invoice reference.
2. Click the Update Payment action. This is available to Admin and Accounts users.



3. Choose Payment Status: Pending, Paid, Overdue, or Cancelled.
4. Enter Payment Date when relevant.
5. Click Save Payment. When status becomes Paid, Biomed users receive a payment-received notification.

### **How To Export Credit Movement**

1. Apply any search or filter needed on Credit Movement.
2. Click Export Excel to download the filtered ledger as a spreadsheet.
3. Click Export PDF to open a printable PDF-style export.
4. Use the export when sharing a period summary with management, finance, or operations.

### **How To Use WhatsApp Summary**

1. Find the transaction row.
2. Click the WhatsApp action.
3. The message uses client, device serial number, model, as-per date, current credit, and floor credit.
4. Review the generated message before sending it externally.

### **How To Review Transaction History**

1. Click the History action on a transaction.
2. Review each entry by action type, changed user, changed date/time, and changed fields.
3. Use the before and after values to confirm whether balance, invoice, payment, activity, contact, image, or notes changed.

### **How To Delete A Transaction**

1. Sign in using the primary admin account configured for deletion access.
2. Find the transaction and click Delete.
3. Confirm the deletion only after checking the transaction history and balance effect.
4. When deleted, the system removes related activity/history rows, restores the contract balance by reversing the transaction effect, and removes the uploaded device image if one exists.

## **3. Purchase Schedule**

**Purpose:** Purchase Schedule shows expected credit-purchase cycles for a device. It is generated from the contract start date or first top-up date, then repeated using the contract top-up frequency.

### **Information Displayed**

- Year and due month.
- Top-up frequency such as every month or every two months.
- Model and Device SN.
- Credit Purchase indicator, invoice details, and payment status.

## How To View A Device Schedule

1. Open Credit Monitoring > Purchase Schedule.
2. Choose a device from the Device dropdown.
3. Review the rolling monthly schedule table.
4. Check whether a cycle is marked Purchased. A cycle is treated as purchased only when a matching top-up has an invoice reference and Paid status.
5. Click Clear to remove the device selection.

## Reminder Logic

- The reminder window is controlled by Settings > General > Schedule Reminder days before cycle.
- Unpurchased cycles within the reminder window can create schedule-reminder notifications.
- Schedule reminders are skipped when the system has already sent the same reminder for the same contract and due date.

## 4. Clients

**Purpose:** Clients lists customer records and summarizes the number of assigned devices, total current credit, low-balance devices, and active/inactive status.

### Information Displayed

- Client name, telephone, and address.
- Device count and total credit balance across active contracts.
- Active or inactive client status.
- Last updated date.
- Actions for generating a client report or viewing the client's devices.

### How To Find A Client

1. Open Clients from the sidebar.
2. Search by client name, address, or telephone.
3. Click Clear to reset the search.
4. Use the Devices action to jump to the Devices tab filtered for that client.

### How To Generate A Client Report

1. Click the Report action for the client.
2. Choose a preset period such as Today, Last 7 days, This month, or Last month.
3. For a custom period, select the start date and then the end date.
4. Click Generate Report.
5. Review the client report summary: total devices, transactions, credit utilization, top-ups, and current credit.
6. Review Device Credit Summary and Transaction Detail.
7. Click Print / Save PDF when a formal copy is needed.

## 5. Devices

**Purpose:** Devices is the operational balance-monitoring tab. It shows every system device, its active contract status, current credit, top-up frequency, and low-balance threshold.

### Information Displayed

- Device serial number and model.
- Assigned client and department.
- Current credit and top-up frequency.
- Low Balance Threshold and status: Active, Low Balance, or Unassigned.
- Last Updated date and threshold edit action for Admin/Biomed users.

### How To Monitor Low-Balance Devices

1. Open Devices from the sidebar.
2. Use Status filter and select Low Balance.
3. Review current credit against the threshold.
4. Open Credit Movement if a top-up, utilization correction, or transaction review is required.
5. Use the Clients filter when you need to focus on one customer.

### How To Edit A Device Threshold

1. Sign in as Admin or Biomed.
2. Open Devices.
3. Find the active contracted device.
4. Click the Edit action in the row.
5. Enter the Low Balance Threshold as zero or a positive number.
6. Save. The Devices and Dashboard low-balance views use the updated threshold.

### How To Import Or Sync Biomed Device Data

1. Start from New Transaction when the device is not already available in the system.
2. Use the device selector to load Biomed devices.
3. Select the Biomed device. The system reads device, client, CRM ID, model, contract, frequency, start date, and commitment data when available.
4. If the Biomed device has a CRM ID, the system can create or update the local client and contract linkage.
5. Continue the new transaction workflow and save.
6. For an existing contract, use the sync action when available to refresh local contract details from Biomed and CRM.

## 6. Revenue Report

**Purpose:** Revenue Report is the Accounts-facing financial review page. It summarizes invoiced credit sales, revenue, usage, and chargeable transactions that are still missing an invoice reference.

## Information Displayed

- Total credits sold for transactions with invoice references and non-cancelled payment status.
- Total revenue based on billed credit, price per credit, and discount.
- Outstanding quotations count and value for chargeable transactions with no invoice reference.
- Average monthly usage based on the selected report period.
- Top customers by credit purchase.
- Outstanding quotation detail with date, customer, device, credits, and quotation value.

## How To Generate A Revenue Report

1. Sign in as Admin or Accounts.
2. Open Revenue Report.
3. Click the Reporting period control.
4. Choose a preset period or select a custom start and end date.
5. Click Generate Report.
6. Review totals first, then check Top customers by credit purchase.
7. Review Outstanding quotation detail and open any row that needs an invoice reference.
8. Click Print / Save PDF to create a report copy.

## Finance Checks

- A transaction must have an invoice reference to count as sold credits and revenue.
- Cancelled transactions are excluded from sold revenue and outstanding quotations.
- A chargeable transaction with Total Charge greater than zero and no invoice reference appears as an outstanding quotation.

## 7. Users

**Purpose:** Users is the admin-only tab for staff account management. It controls who can sign in and which system areas they can access.

### Information Displayed

- Full name, username, email, role, active/inactive status, and last login.
- Search by name, username, or email.
- Filters for status and role.
- Actions to activate/deactivate or edit a user.

### How To Add A User

1. Sign in as Admin.
2. Open Users and click Add User.
3. Enter Full Name, Username, Email, Role, and Password.
4. Click Create User.
5. Tell the user to sign in and change their password from Profile if needed.

## How To Edit A User

1. Open Users.
2. Search or filter until you find the account.
3. Click Edit.
4. Update full name, email, role, active status, or enter a new password.
5. Leave New Password blank when the current password should stay unchanged.
6. Click Save Changes.

## How To Activate Or Deactivate A User

1. Open Users.
2. Find the account.
3. Click the activate/deactivate action.
4. Confirm the action. Deactivated users cannot continue normal access.
5. The signed-in user cannot deactivate their own account from the user list.

## 8. Settings

**Purpose:** Settings controls system defaults and reference data used by transaction entry and notifications.

### Settings Tabs

| Settings tab        | Purpose                                       | Important fields/actions                                                                         |
|---------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------|
| General             | Default operational settings.                 | Default Low Balance Threshold and Schedule Reminder days before cycle.                           |
| Activities          | Activity options used inside Credit Movement. | Add Activity, edit activity name/status, set balance effect as increase, decrease, or neutral.   |
| Email Notifications | External mailing list for alert emails.       | Add recipient, edit email/name/active status, delete recipient. Visible only to admin@admin.com. |

### How To Update General Settings

1. Sign in as Admin.
2. Open Settings > General.
3. Set Default Low Balance Threshold. This value is used for newly imported devices unless overridden.
4. Set Schedule Reminder days before cycle. This controls how early schedule reminder alerts are raised.
5. Click Save Settings.

### How To Add An Activity

1. Open Settings > Activities.
2. Click Add Activity.
3. Enter the activity name.
4. Choose the balance effect: increase, decrease, or neutral.
5. Keep Active checked if the option should appear in transaction forms.

6. Click Create Activity.

## **How To Edit Or Deactivate An Activity**

1. Open Settings > Activities.
2. Search or filter by status or balance effect.
3. Click Edit to change the name, balance effect, or active status.
4. Click Save Activity.
5. Use Deactivate when the activity should no longer be available for new transaction entry.

## **How To Manage Email Recipients**

1. Sign in using admin@admin.com.
2. Open Settings > Email Notifications.
3. Click Add Recipient.
4. Select an existing user or manually enter email and name.
5. Click Add Recipient.
6. Back on the mailing list, update email, name, or active status and click Save when needed.
7. Click Delete to remove a recipient from the mailing list.

## **9. Profile And Notifications**

**Purpose:** Profile is for the signed-in user's own account details and password. Notifications appear in the dashboard/header areas and support operational follow-up.

### **How To Change Password**

1. Open the profile menu at the bottom of the sidebar.
2. Click Profile.
3. Enter Current Password.
4. Enter New Password.
5. Enter Confirm Password.
6. Click Update Password.

### **Notification Types**

- Low Credit Balance Alert: raised when a contract balance is at or below its threshold.
- Payment Received: sent to Biomed when Accounts marks a transaction payment status as Paid.
- Credit Purchase Reminder: raised for upcoming unpaid purchase cycles based on schedule reminder settings.
- General: reserved for general system notices.

### **How To Act On Notifications**

1. Read the notification title and message.
2. For low balance, open Devices or Credit Movement and check the related contract/device.
3. For payment received, open Credit Movement and confirm the invoice/payment details.

4. For purchase reminder, open Purchase Schedule and check whether a top-up has been purchased and paid.
5. Mark notifications as read after follow-up is complete.

## **System Controls And Good Practice**

### **Required Fields And Validation**

- New or edited transactions require Device, Activity, Date From, and Date To.
- Account numeric fields must be zero or more; Discount must be between 0 and 100.
- Low-balance threshold must be zero or more.
- Email notification recipients require a valid email address.
- A transaction can contain multiple activities, but only one top-up activity.

### **Recommended Operating Routine**

1. Start each day on Dashboard and review low-balance and schedule alerts.
2. Open Devices to confirm affected serial numbers and thresholds.
3. Open Credit Movement to record usage, losses, top-ups, or finance updates.
4. For paid top-ups, ensure invoice reference, payment status, and payment date are filled.
5. Use Clients or Revenue Report for formal summaries.
6. Use History before changing or deleting sensitive transaction records.

### **Complete Information Flow**

**Credit system flow:** Biomed/CRM device data -> Local client/department/device/contract -> Credit transaction -> Contract balance -> Low-balance and schedule checks -> Accounts invoice/payment update -> Client and revenue reports -> Dashboard KPIs and notifications